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ICON Central Office - Madhapur - Hyderabad

SEC: Sr. Elite & Target(C-120, C-IPL & IPL-IC)

JEE-MAIN

Date: 08-10-2023

Time: 09.30AM to 12.30PM

CTM-04

Max. Marks: 300

IMPORTANT INSTRUCTION:

1. Immediately fill in the Admission number on this page of the Test Booklet with **Blue/Black Ball Point Pen** only.
2. The candidates should not write their Admission Number anywhere (except in the specified space) on the Test Booklet/ Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of 90 questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and subject having **two sections**.

(I) Section –I contains 20 **multiple choice** questions with only one correct option.

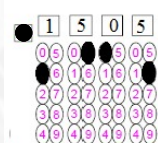
Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only, if more than 5 questions attempted, First 5 attempted questions will be considered.

- The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

To cancel any attempted question bubble on the question number box.

For example: To cancel attempted question 21. Bubble on 21 as shown below



Question Answered for Marking **Question Cancelled for Marking**

Marking scheme: +4 for correct answer, 0 if **not attempt** and -1 in all other cases.

6. Use **Blue / Black Point Pen only** for writing particulars / marking responses on the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electron device etc, except the Identity Card inside the examination hall.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capital): _____

Admission Number:

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Candidate's Signature: _____

Invigilator's Signature: _____

08-10-2023_Sr.Elite & Target(C-120, C-IPL & IPL-IC)_ Jee-Main_CTM_04_Test Syllabus

PHYSICS : W-1 TO 8 Syllabus

CHEMISTRY : W-1 TO 8 Syllabus

MATHEMATICS : W-1 TO 8 Syllabus



PHYSICS

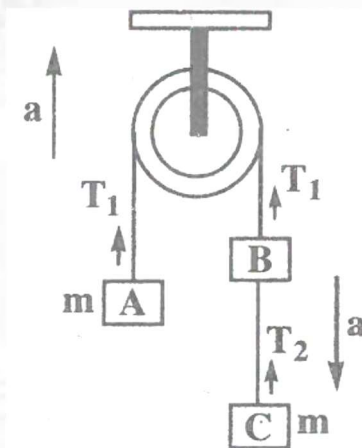
Max Marks: 100

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

- An experiment measures quantities a , b , c and X as $X = \frac{ab^2}{c^3}$; if the percentage errors in a , b , c are $+1\%$; $+3\%$; $+2\%$ respectively, the percentage error in X is
A) $\pm 13\%$ B) $\pm 7\%$ C) $\pm 4\%$ D) $\pm 1\%$
- A stone is thrown vertically up with speed v_0 from a cliff of height H . Find the average speed of stone till the moment it reaches ground. Given $H=100\text{ m}$, $v_0=10\text{ m/s}$, $g=10\text{ m/s}^2$
A) $\frac{64}{1+\sqrt{21}}\text{ m/s}$ B) $\frac{110}{1+\sqrt{21}}\text{ m/s}$ C) 55 m/s D) $110(1+\sqrt{21})\text{ m/s}$
- Three equal weights of mass m each are hanging on a string passing over a fixed pulley as shown in fig. What are the tensions in the string connecting weight A to B and B to C?



- A) $7/4\text{ mg}$; $3/4\text{ mg}$ B) $7/3\text{ mg}$; $4/3\text{ mg}$
C) $4/3\text{ mg}$; $2/3\text{ mg}$ D) $3/4\text{ mg}$; $mg/4$

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JEE MAIN
2023

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AIR NO. 23052 (2023)
sri chaitanya
6th-12th Class

300
300
MARKS



RANK
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JEE Advanced
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341
360
MARKS



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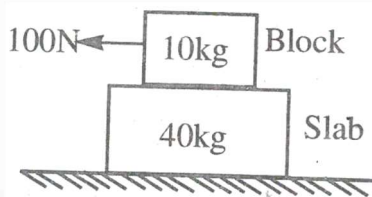
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MARKS



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4. A 40kg slab rests on a frictionless floor. A 10 kg block rests on top of the slab. The coefficient kinetic friction between the block and the slab is 0.40. A horizontal force of 100N is applied on the 10kg block Find the resulting acceleration of the slab. ($g=10\text{ms}^{-2}$)



- A) 3ms^{-2} B) 4ms^{-2} C) 1ms^{-2} D) 2ms^{-2}
5. A rotor of radius (r) is rotating about its own vertical axis and a person in contact with inner wall of rotor remains in equilibrium without slipping down. If (ω) is angular velocity of rotor and ' μ ' is minimum coefficient of friction between person and the wall of rotor. Then which of the following is correct
- 1) $\mu \propto \omega^2$ 2) $\mu \propto \frac{1}{r}$ 3) $\mu \propto \frac{1}{\omega^2}$ 4) $\mu \propto r$
- A) 1,2 true B) 1,4 true C) 2,3 true D) 3,4 true
6. A constant force acting on a body of mass 1 kg provides it a kinetic energy of 1800 J by the end of 5th second. If the body was initially at rest then magnitude of force is equal to
- A) 6N B) 10 N C) 5N D) 12N
7. A closed system consisting of ideal gas undergoes isothermal process. Heat energy is supplied to the system, choose the option showing correct statements
- 1) Internal energy increases 2) Internal energy remains constant
3) Work done by system is negative 4) Internal energy decrease
5) work done by system is positive
- A) 1,3,5 B) 2,5 C) 1,2,3,4 D) 2,3

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MARKS

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1



8. A ball of mass 0.4 kg moving with a uniform speed of 2ms^{-1} strikes normally a wall and rebound. Assuming the collision to be elastic and the time of contact of the wall as 0.4s , find the force exerted on the ball.
- A) 1 N B) 2 N C) 3 N D) 4 N
9. A solid cylinder starts pure rolling from top of a fixed inclined plane of slant length 60cm and inclination 30° . Velocity of centre of mass when the cylinder reaches the bottom is equal to ($g = 10\text{m/s}^2$)
- A) 1 m/s B) 3 m/s C) 2 m/s D) 4 m/s
10. Three particles of masses 1 kg , 2kg and 3kg are placed at the vertices A, B and C of an equilateral triangle ABC. If A and B lie at $(0,0)$ and $(1, 0) \text{ m}$, the Co-ordinates of their centre of mass are
- A) $\left(\frac{\sqrt{3}}{2}\text{m and } \frac{7}{6}\text{m}\right)$ B) $\left(\frac{7}{6}\text{m and } \frac{\sqrt{3}}{4}\text{m}\right)$ C) $\left(\frac{7}{12}\text{m and } \frac{\sqrt{3}}{4}\text{m}\right)$ D) $\left(\frac{7}{12}\text{m and } \frac{7}{12}\text{m}\right)$
11. Two balls are projected with equal speed at an angle of 30° and the other at 60° with horizontal. Find the ratio of maximum heights of both the balls
- A) $\frac{1}{4}$ B) $\frac{3}{1}$ C) $\frac{1}{5}$ D) $\frac{1}{3}$
12. Let I_{cm} be moment of inertia of disc passing through centre and perpendicular to its plane. I_{AB} be moment of inertia about axis AB that is in the plane of disc and at a distance of $\frac{2R}{3}$ from centre. Find $\frac{I_{cm}}{I_{AB}}$
- A) $\frac{18}{25}$ B) $\frac{1}{4}$ C) $\frac{9}{17}$ D) $\frac{1}{2}$

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MARKS**RANK**
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6th-12th Class**720**
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13. A point mass m and a uniform thin rod (mass M and length L) are shown in figure. Find the magnitude of gravitational force attraction between them



- A) $\frac{GMm}{ah}$ B) $\frac{GMm}{a(a+L)}$ C) $\frac{GMm}{(a+L)^2}$ D) $\frac{4GMm}{(2a+L)^2}$
14. Column -I few physical quantities and column -II their dimensional formulae given

	Column-I		Column-II
P)	Pressure gradient	M)	$[M^1 L^2 T^{-2}]$
Q)	Energy density	N)	$[M^1 L^1 T^{-1}]$
R)	Torque	O)	$[M^1 L^{-2} T^{-2}]$
S)	Impulse	T)	$[M^1 L^{-1} T^{-2}]$

- A) P-O, Q-M, R-N S-T B) P-O, Q-T, R-M, S-N
C) P-M, Q-T, R-N, S-O D) P-M, Q-O, R-N, S-T
15. A mass ' M ' is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes SHM of time period ' T '. If the mass is increased by ' m ', time period becomes $\frac{5T}{3}$ then the ratio of (m/M) is
- A) $\frac{3}{2}$ B) $\frac{25}{9}$ C) $\frac{16}{9}$ D) $\frac{5}{3}$





16. A light rod of cross-sectional area A and Young's modulus Y is arranged as shown. The applied force $F=250\text{N}$, length of rod is 1 m . Find the extension in rod.

Given that: $A = 6.25 \times 10^{-4} \text{ m}^2$, $Y = 10^{10} \text{ N / m}^2$



- A) $5 \times 10^{-6} \text{ m}$ B) $4 \times 10^{-5} \text{ m}$ C) $2 \times 10^{-6} \text{ m}$ D) $4 \times 10^{-7} \text{ m}$
17. The rate of steady volume flow of water through a capillary tube of length ' l ' and radius r under a pressure difference of P is V . This tube is connected with another tube of the same length but half the radius in series then the rate of steady volume flow through them under same pressure difference across the combination is
- A) $\frac{V}{16}$ B) $\frac{V}{17}$ C) $\frac{16V}{17}$ D) $\frac{17V}{16}$
18. The reading of a spring balance when a block is suspended from it in air is 60 N . This reading is changed to 40 N when the block is submerged in water. The specific gravity of block must be therefore
- A) 3 B) 2 C) 6 D) $3/2$
19. Chose the option showing the correct relation between Poisson's ratio (σ) bulk modulus (B) and modulus of rigidity (G)
- A) $\sigma = \frac{3B-2G}{2G+6B}$ B) $\sigma = \frac{6B+2G}{3B-2G}$ C) $\sigma = \frac{9BG}{3B+G}$ D) $B = \frac{3\sigma-3G}{6\sigma+2G}$



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MARKS**



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20. Assertion: If the weight of the lift is equal to the tension force of the cable wire then it moves with uniform velocity
Reason: If the lift moves downward with an acceleration, then the contact force between the boy's feet and lift floor is more than the weight of boy
- A) Both are true and reason is correct explanation of assertion
B) Both are true and reason not correct explanation of assertion
C) Assertion is true reason is false
D) Reason is true, assertion is false

(NUMERICAL VALUE TYPE)

Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if **not attempt** and -1 in all other cases.

21. A ladder 'AB' of weight 300 N and length 5m is lying on a horizontal surface. Its centre of gravity is at a distance of '2m' from end A. A weight of 80N is attached at end B. The work done in raising the ladder to the vertical position with end 'A' in contact with the ground is (in J)
22. Two particles of masses 2 m are at a distance 'd' apart. Under their mutual gravitational force, they start moving towards each other. The acceleration of their centre of mass when they are $d/2$ apart is _____
23. Gravitational acceleration on the surface of the planet is $\frac{\sqrt{6}}{11}g$, where g is the acceleration due to gravity on the surface of earth. The average mass density of the planet is $2/3$ times that of the earth. If the escape speed on the surface of the earth is taken to be 11 km.s^{-1} then find the escape speed on the surface of the planet (in km.s^{-1})

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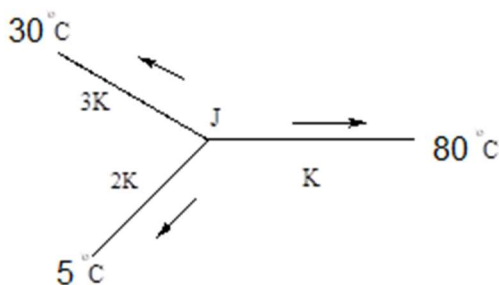
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MARKS



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24. Several spherical drops of radius r of a liquid of surface tension T coalesce to form a single drop of radius R and volume V . The energy released in the process is $nVT\left[\frac{1}{r}-\frac{1}{R}\right]$, where $n =$
25. Time period of rotation of a planet is 24 hours. If radius decreases to $\frac{1}{4}$ th of original value then the new time period is y hours. Find $4y$
26. The temperature for which the Celsius and Fahrenheit scales show the same reading but with opposite sign is $11.43n$ where n is
27. Three rods with thermal conductivities K , $3K$ and $2K$ have same dimensions as shown in figure. The temperature of the junction in the steady state is _____°C.



28. Find change in kinetic energy (in J) if a body is displaced from $(1, 2)m$ to $(2, 3)m$ on applying a force of $\vec{F} = 4x^2\hat{i} + 3y^3\hat{j} N$.
29. A particle is executing S H M with an amplitude of A along X-axis. Origin is taken as mean position. The position where kinetic energy and potential energy of particle are same is $\frac{A}{\sqrt{P}}$. The value of P is
30. Work done in increasing the size of a soap bubble from a radius of 3 cm to 5 cm is $x \times \pi (in \mu J)$. Find x (given surface tension of soap solution $3 \times 10^{-2} Nm^{-1}$)

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MARKS



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MARKS



RANK 1

**CHEMISTRY****Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

31. Which one of the following species is planar and non-polar with two lone pairs of electrons on the central atom?
- A) ClF_3 B) XeF_5^- C) PCl_5 D) BrF_5
32. Consider the following statements and arrange in the order of true/false as given in the codes
- S_1 : concentrated solution of an alkali metal in liquid ammonia is diamagnetic
- S_2 : Increasing stability of the peroxide and super oxide of alkaline earth metal down the group, is due to the stabilization of large anions by large cations through lattice energy effects.
- S_3 : In the vapour phase BeCl_2 tends to form a chloro-bridged dimer.
- S_4 : Beryllium hydroxide can react with acid and alkali
- A) T T T T B) F F F F C) T F T F D) F T F T
33. statement-1: Aqueous solution of hydrogen peroxide is kept in glass or metal container containing some urea or phosphoric acid.
- Statement-2: Urea or phosphoric acid as a negative catalyst for the decomposition of hydrogen peroxide.
- A) Statement 1 and statement 2 are correct and statement 2 is the correct explanation of statement 1.
- B) Statement 1 and statement 2 are correct but statement 2 is not correct explanation of statement 1.
- C) Statement 1 is correct but statement 2 is false.
- D) Statement 1 is false but statement 2 is correct.

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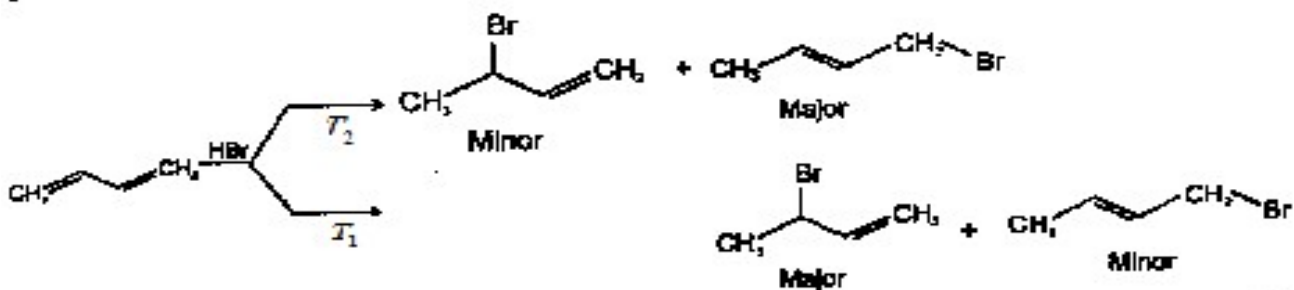
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34. Electrons in a cathode ray tube have been emitted with a velocity of 1000ms^{-1} . The number of following statements which is/are true about the emitted radiation is _____. Given:
 $h = 6 \times 10^{-34} \text{ Js}$, $m_e = 9 \times 10^{-31} \text{ kg}$.
- 1) The deBroglie wavelength of the electron emitted is 666.67 nm.
 - 2) The characteristic of electrons emitted depend upon the material of the electrodes of the cathode ray tube.
 - 3) The cathode rays start from cathode and move toward anode.
 - 4) The nature of the emitted electrons depends on the nature of the gas present in cathode
- A) 1 & 3 B) 2 & 4 C) 1 & 4 D) All
35. Given below are two statements:
Statement-I: According to Bohr's model of hydrogen atom, the angular momentum of an electron in a given stationary state is quantized
Statement-II: The concept of electron in Bohr's orbit violates the Heisenberg uncertainty principle In the light of the above statements; choose the most appropriate answer from the options given below
- A) Statement-I is correct but Statement-II is incorrect
 - B) Both Statement-I and Statement-II are incorrect
 - C) Both Statement-I and Statement II are correct
 - D) Statement I is incorrect but Statement-II is Correct
36. Which of the following statements are not correct?
- 1) The electron gain enthalpy of **F** is more negative than that of **Cl**.
 - 2) Ionization enthalpy decreases in a group of periodic table.
 - 3) The electro negativity of an atom depends upon the atoms bonded to it.
 - 4) Al_2O_3 and NO are examples of amphoteric Oxides.
- A) 1, 3 & 4 B) 1, 2 & 3 C) 3 only D) All





37.

Here T_1 & T_2 are two different temperatures

Statement 1: Low temperature favours formation of 1, 2 - addition product

Statement 2: High temperature favours formation of 1, 4 - addition product

A) Statement 1 is True & Statement 2 also True

B) Both the statements are false

C) Statement 1 is True, Statement 2 is False.

D) Statement 1 is False, Statement 2 is True.

38. Which one of the following is the correct statement?

A) Boric acid is a protonic acid

B) Both Tl^{3+} and Al^{3+} ions act as oxidising agent in aqueous solution.

C) Hydrogen bonding in H_3BO_3 gives it a layered structure.

D) $B(OEt)_3$ imparts blue colour to the burner flame.

39. In which of the following processes, the bond order increases and paramagnetic character changes to diamagnetic one?

1) $NO \rightarrow NO^+$ 2) $O_2 \rightarrow O_2^-$ 3) $N_2 \rightarrow N_2^+$ 4) $O_2 \rightarrow O_2^+$

A) 2

B) 3

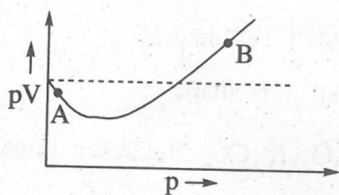
C) 1

D) 4





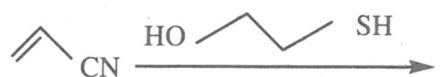
40. For 1 mol of gas. the plot of pV vs. p is shown below. P is the pressure and V is the volume of the gas

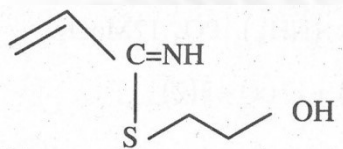
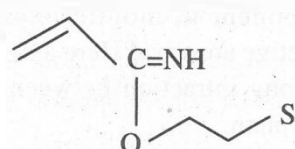
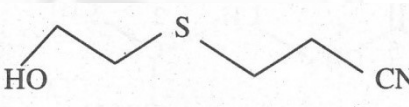
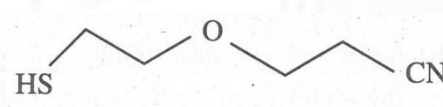


What is the value of compressibility factor at point A ?

- 1) $1 + \frac{b}{v}$ 2) $1 + \frac{a}{RTV}$ 3) $1 - \frac{a}{RTV}$ 4) $1 - \frac{b}{v}$
A) 1 B) 2 C) 3 D) 4

41. The major product for the following reaction is



- 1)  2) 
3)  4) 
A) 1 B) 3 C) 2 D) 4



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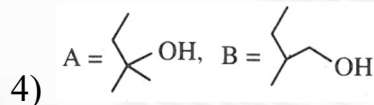
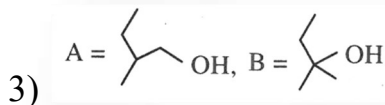
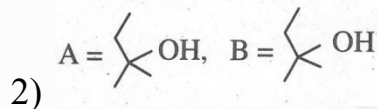
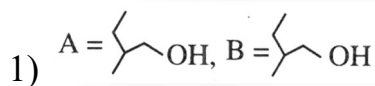
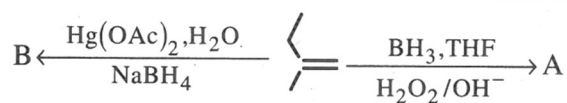
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42. Find out the major products from the following reactions.



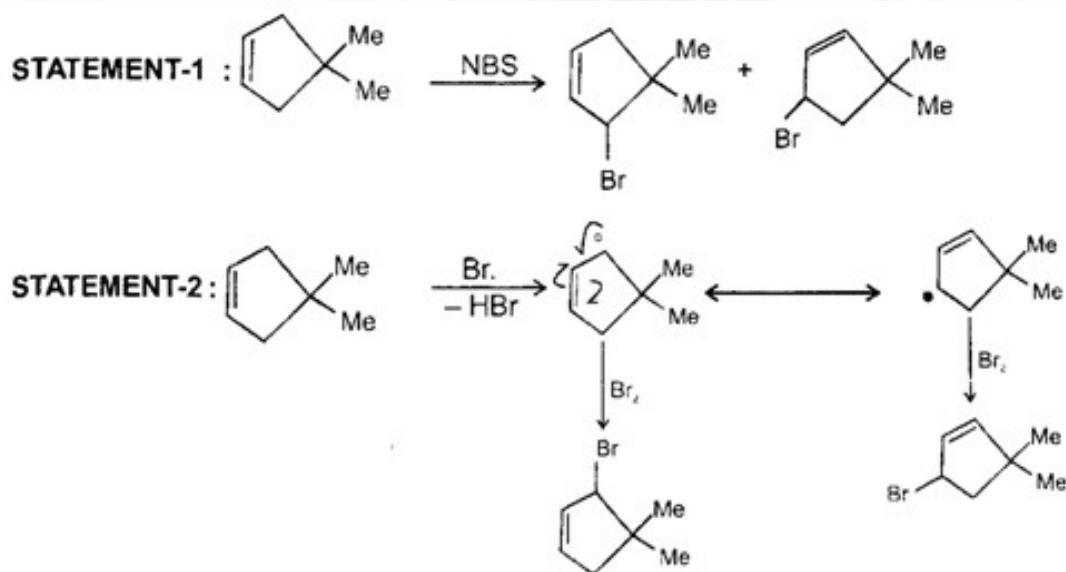
A) 1

B) 3

C) 4

D) 2

43.



A) Statement 1 is True, Statement 2 is True; Statement 2 is a correct explanation for statement 1.

B) Statement 1 is True, Statement 2 is True; Statement 2 is NOT a correct explanation for statement 1.

C) Statement 1 is True, Statement 2 is False.

D) Statement 1 is False, Statement 2 is True.

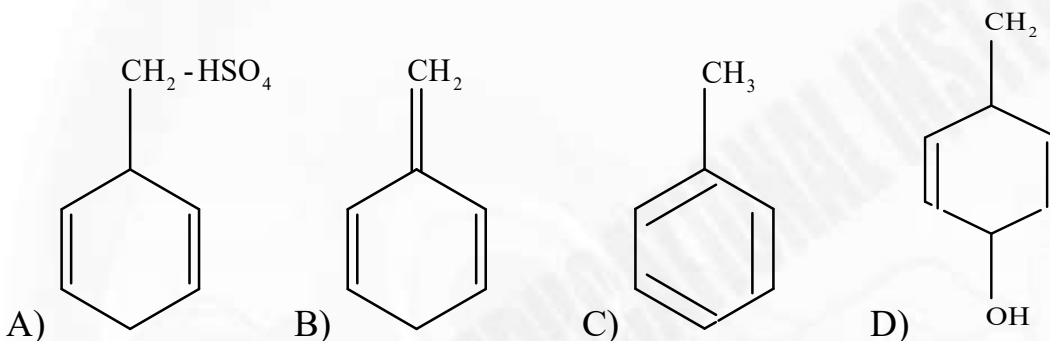
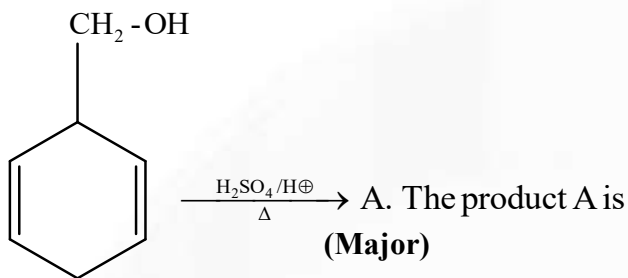
Sec: Sr.Elite & Target(C-120, C-IPL & IPL-IC)

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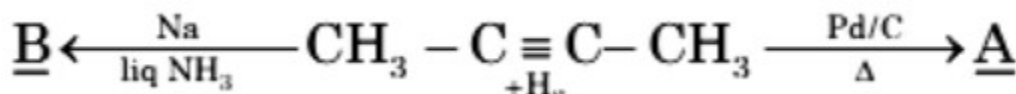




44.



45. But-2-yne is reacted separately with one mole of Hydrogen as shown below:

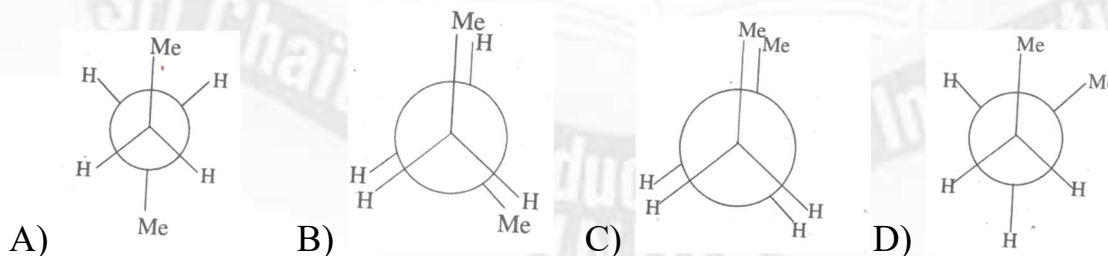


- A) A is more soluble than B.
 B) The boiling point & melting point of A are higher and lower than B respectively.
 C) A is more polar than B because dipole moment of A is Zero.
 D) Adds easily to B than A.

Identify the incorrect statement from the option given below:

- A) A, C & D only B) A and B only C) B, C & D only D) B and C only

46. Which of the following conformation will be the most stable?


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47. Given below are two statements, one is labelled as Assertion A and the other is labelled as Reason R.
Assertion A: Beryllium has less negative value of reduction potential compared to the other alkaline earth metals.
Reason R: Beryllium has large hydration energy due to small size of Be^{2+} but relatively large value of atomization enthalpy
In the light of the above statements, choose the most appropriate answer from the options given below
A) A is correct but R is not correct
B) A is not correct but R is correct
C) Both A and R are correct but R is NOT the correct explanation of A
D) Both A and R are correct and R is the correct explanation of A
48. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): Acetal / Ketal is stable in basic medium.
Reason (R) : The high leaving tendency of alkoxide ion gives the stability to acetal ketal in basic medium.
In the light of the above statements, choose the correct answer from the options given below
A) Both A and R are true but R is NOT the correct explanation of A
B) A is false but R is true
C) A is true but R is false
D) Both A and R are true and R is the correct explanation of A
49. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R
Assertion (A): Physical properties of isotopes of hydrogen are different.
Reason (R): Mass difference between isotopes of hydrogen is very large
In the light of the above statements, choose the correct answer from the options given below:
A) Both A and R are true but R is NOT the correct explanation of A
B) A is false but R is true
C) Both A and R are true and R is the correct explanation of A
D) A is true but R is false

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50. **Statement-1** : Bond order in a molecule can assume any value positive or negative, integral or fractional when number of electrons in bonding molecular orbitals is greater than that in anti-bonding orbitals.

Statement-2 : Bond order depends on the number of electrons in the bonding and anti-bonding orbitals.

A) Statement 1 is True, Statement 2 is True; Statement 2 is a correct explanation for statement 1.

B) Statement 1 is True, Statement 2 is True; Statement 2 is NOT a correct explanation for statement 1.

C) Statement 1 is True, Statement 2 is False.

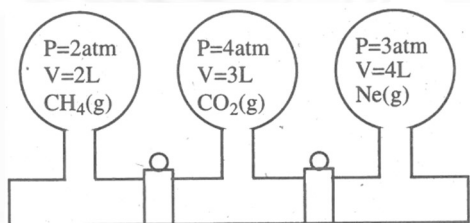
D) Statement 1 is False, Statement 2 is True.

(NUMERICAL VALUE TYPE)

Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only. First 5 attempted questions will be considered if more than 5 questions attempted. The Answer should be within **0 to 9999**. If the Answer is in **Decimal** then round off to the **nearest Integer** value (Example i.e. If answer is above **10** and less than **10.5** round off is **10** and If answer is from **10.5** and less than **11** round off is **11**).

Marking scheme: +4 for correct answer, 0 if **not attempt** and -1 in all other cases.

51. In H sample, e^- is present in the orbit in which angular momentum of e^- is $\frac{2h}{\pi}$. Electron makes transition upto ground state. If 3rd highest energy photon has energy of X e V then approximate numerical value of $\left(\frac{2x}{5}\right)$ is



52.

Three bulbs are filled with CH_4 , CO and Ne as shown in the picture. The bulbs are connected through pipes of zero volume. When the stopcocks are opened and the temperature is kept constant throughout, the pressure of the system is found to be atm. (Nearest Integer))



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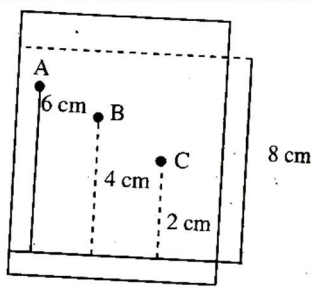
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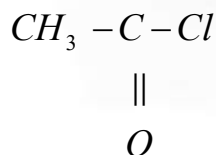
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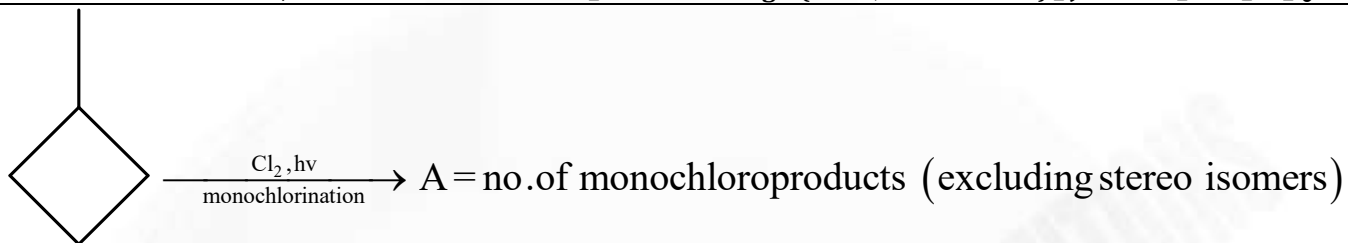


53. An element forms a hydride containing 5.928% hydrogen. 0.1794g of the gaseous hydride occupies 118 mL at STP. If equivalent weight of element is X and molecular weight of the hydride is Y ; then report $s \frac{X+Y}{10}$ to the closest integer
54. Two acid solutions A and B are titrated separately each with 25 mL of 1N Na_2CO_3 solution. The volumes of each acid used for titration are 10 ml and 40 ml respectively. Find out the volume ratio of acid B and A respectively, which should be mixed to prepare 1 mL of 1N solution.
55. The ion A^{n+} is oxidized to AO_3^- by MnO_4^- changing to Mn^{+2} in acidic medium. Given that 2.68×10^{-3} mole of A^{n+} requires 1.61×10^{-3} mole of MnO_4^- . Thus, n is
56. Three organic compounds A, B and C were allowed to run in thin layer chromatography using hexane and gave the following result (see figure). The R_f value of the most polar compound is _____ $\times 10^{-2}$ cm



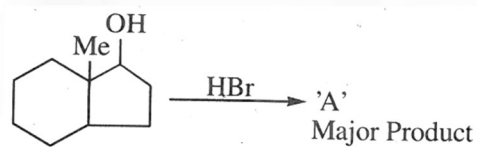
57. How many molar equivalents are $R-MgX$ is required for the reaction with

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58.

59. On heating a mixture of 8 moles of Li_2CO_3 and KHCO_3 in 3: 1 ratio, the number of moles of CO_2 liberated is



60.

The number of hyper conjugation structures involved to stabilize carbocation formed in the above reaction is _____.

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**MATHEMATICS****Max Marks: 100****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 in all other cases.

61. Let $S = \{1, 2, 3, \dots, 50\}$. The number of non-empty subsets A of S such that the product of elements in A is even is:
- A) $2^{50} - 1$ B) $2^{25} (2^{25} - 1)$ C) $2^{50} + 10$ D) $2^{50} + 1$
62. Let $R = \{(x, y) : x, y \in N \text{ and } x^2 - 4xy + 3y^2 = 0\}$, where N is the set of all natural numbers. Then the relation R is
- A) Reflexive but neither symmetric nor transitive B) Symmetric
C) Reflexive and symmetric D) Reflexive and transitive
63. If A is order 3 square matrix such that $|A| = 2$, then $|adj(adj(adj A))| = ______$ is
- A) 512 B) 256 C) 64 D) 32
64. Elements of a matrix A of order 10×10 are defined as $a_{ij} = \omega^{i+j}$ (where ω is imaginary cube root of unity), then trace (A) of the matrix is
- A) 0 B) ω^3 C) 3 D) ω^2
65. If the domain of the function $f(x) = \frac{\cos^{-1} \sqrt{x^2 - x + 1}}{\sqrt{\sin^{-1} \left(\frac{2x-1}{2} \right)}}$ is the interval $(a, b]$ then $a+b$ is equal to :
- A) $\frac{3}{2}$ B) 2 C) $\frac{1}{2}$ D) 1
66. If $\Delta(x) = \begin{vmatrix} \tan x & \tan(x+h) & \tan(x+2h) \\ \tan(x+2h) & \tan x & \tan(x+h) \\ \tan(x+h) & \tan(x+2h) & \tan x \end{vmatrix}$, then the value of $\lim_{h \rightarrow 0} \frac{\Delta(\pi/3)}{\sqrt{3}h^2}$ is
- A) 144 B) 81 C) 64 D) 36

Sec: Sr.Elite & Target(C-120, C-IPL & IPL-IC)

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67. Let the volume of a parallelepiped whose coterminal edges are given by $\vec{u} = \hat{i} + \hat{j} + \lambda \hat{k}$, $\vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. Unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$, then $\cos \theta$ can be:
- A) $\frac{7}{\sqrt{3}}$ B) $\frac{7}{6\sqrt{3}}$ C) $\frac{5}{7}$ D) $\frac{5}{3\sqrt{3}}$
68. Let two vertices of a triangle ABC be (2,4,6) and (0,-2,-5), and its centroid be (2,1,-1). If the image of the third vertex in the plane $x + 2y + 4z = 11$ is (α, β, γ) , then $\alpha + \beta + \gamma$ is ____
- A) 12 B) 14 C) 10 D) 15
69. The distance of the point (1, -2, 3) from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are 2, 3, -6 is :
- A) 3 B) 5 C) 2 D) 1
70. Let $a, b \in \mathbb{R}$ be such that the function f given by $f(x) = \log|x| + bx^2 + ax, x \neq 0$ has extreme values at $x = -1$ and $x = 2$.
Statement I: f has local maximum at $x = -1$ and $x = 2$.
Statement II: $a = 1/2$ and $b = -1/4$
A) I is false and II is true.
B) Both I and II are true and II is the correct explanation of I
C) Both I and II are true and II is not the correct explanation of I
D) I is true and II is false.
71. The value of $\lim_{x \rightarrow 0} \left(\left[\frac{100x}{\sin x} \right] + \left[\frac{99 \sin x}{x} \right] \right)$ (where $[.]$ represents the greatest integral function) is
- A) 199 B) 198 C) 0 D) 197
72. Let $f: (-5, 5) \rightarrow \mathbb{R}$ be a differentiable function with $f(4) = 1, f'(4) = 1, f(0) = -1$ and $f'(0) = 1$. If $g(x) = (f(2f^2(x) + 2))^2$, then $g'(0)$ equals
- A) 4 B) -4 C) 8 D) -8



73. If $I(x) = \int e^{\sin^2 x} (\cos x \sin 2x - \sin x) dx$ and $I(0) = 1$

Statement I: $I\left(\frac{\pi}{3}\right)$ is equal to $\frac{1}{4}e^{3/4}$

Statement II: $I\left(\frac{\pi}{4}\right)$ is equal to $\frac{1}{\sqrt{2}}(\sqrt{e})$

- A) Statement I is true, statement II is true.
 B) Statement I is false, statement II is true.
 C) Statement I is false, statement II is false.
 D) Statement I is true, statement II is false.

74. Statement 1: The value of the integral $\int_{\frac{\pi}{24}}^{\frac{5\pi}{24}} \frac{dx}{1 + \sqrt{\tan 2x}}$ is equal to $\frac{\pi}{12}$

Statement 2: $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

- A) Statement-1 is true, Statement 2 is true; Statement-2 is a correct explanation for Statement-1
 B) Statement-1 is true; Statement 2 is true; Statement-2 is not a correct explanation for Statement 1.
 C) Statement-1 is true; Statement-2 is false.
 D) Statement-1 is false, Statement-2 is true.

75. If $f(x) = \begin{cases} x^2 - ax + 3, & x \text{ is rational} \\ 2-x, & x \text{ is irrational} \end{cases}$ is continuous at exactly two points, then the possible values of a are

- A) $(2, \infty)$ B) $(-\infty, 3)$ C) $(-\infty, -1) \cup (3, \infty)$ D) None of these

76. The number of positive real solutions of equation $x^7 + 5x^3 + 3x + 1 = 0$ is _____

- A) 0 B) 1 C) 2 D) 3

77. If $\int_0^x f(t) \sin t dt = \text{constant}$, $0 < x < 2\pi$ and $f(\pi) = 2$, then the value of $f(\pi/2)$ is

- A) 3 B) 2 C) 4 D) 8



78. Statement I: Let $y = y(x)$ be the solution of the differential equation

$$x^3 dy + (xy - 1) dx = 0, x > 0, y\left(\frac{1}{2}\right) = 3 - e. \text{ Then } y(1) = 1$$

Statement II: If $\frac{dy}{dx} + y p(x) = Q(x)$ Then solution is $y(I.F) = \int Q(x)(I.F) dx$

A) Statement-1 is true, Statement 2 is true, Statement 2 is not a correct explanation of Statement 1.

B) Statement 1 is false, Statement 2 is true.

C) Statement 1 is true, Statement 2 is false.

D) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

79. Area bounded by $f(x) = \frac{x^2 - 1}{x^2 + 1}$ and the line $y = 1$ is

A) $\pi/2$ sq. units B) π sq. units C) $3\pi/2$ sq. units D) 2π sq. units

80. Given $f'(1) = 1$ and $f(2x) = f(x) \forall x > 0$. If $f'(x)$ is differentiable then there exists a number $c \in (2, 4)$ such that $f''(c)$ equals to

A) $1/4$ B) $-1/2$ C) $-11/4$ D) $-1/8$

(NUMERICAL VALUE TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempt and -1 in all other cases.

81. Let $A = \{a, b, c\}$ and $B = \{1, 2, 3, 4\}$. Then the number of elements in the set

$C = \{f : A \rightarrow B / 3 \in f(A) \text{ and } f \text{ is not one-one}\}$ is _____

82. Let M be any 3×3 matrix with entries from the set $\{0, 1, 2\}$. The maximum number of such matrices, for which the sum of diagonal elements of $M^T M$ is seven is given by K then $\frac{K}{10}$ is equal to _____





83. Let α, β, γ are the real roots of the equation $x^3 + ax^2 + bx + c = 0$ ($a, b, c \in R$ and $a \neq 0$). If the system of equations (in u, v and w) given by
- $$\alpha u + \beta v + \gamma w = 0, \quad \beta u + \gamma v + \alpha w = 0, \quad \gamma u + \alpha v + \beta w = 0$$
- Has non trivial solutions, then the value of $\frac{a^2}{b}$ is _____
84. Let for a triangle ABC, $\overline{AB} = -2\hat{i} + \hat{j} + 3\hat{k}; \overline{CB} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k};$
 $\overline{CA} = 4\hat{i} + 3\hat{j} + \delta\hat{k}$ If $\delta > 0$ and the area of the triangle ABC is $5\sqrt{6}$ then $\frac{1}{15}(\overline{CB} \cdot \overline{CA})$ is equal _____
85. Let $\vec{p}, \vec{q}$ be two vectors. If $\vec{p} = 2\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 2\hat{j} + \hat{k}$ vector $\vec{r} = (\alpha\hat{i} + \beta\hat{j} + \gamma\hat{k})$ is perpendicular to each of the vectors $(\vec{p} + \vec{q})$ and $(\vec{p} - \vec{q})$, and $|\vec{r}| = \sqrt{3}$, then $|\alpha| + |\beta| + |\gamma|$ is equal to _____.
86. The number of points of discontinuity for $f(x) = \text{sgn}(\sin x)$, $x \in [0, 4\pi]$ is _____.
87. If the curves $2x = y^2$ and $2xy = k$, intersect orthogonally, then the value of k^2 - _____.
88. The rate of change in the volume of sphere is equals to the rate of change in its radius. If the radius is $\frac{1}{k\sqrt{\pi}}$ then the value of 'k' is _____.
89. If $f(x) = \int \frac{\cos^3 x}{\sin^2 x + \sin x} dx$ and $f(\pi/2) = -1$ then $f(3\pi/2) =$ _____.
90. Area bounded by the curves $\min. \{|x|, |y|\} = 1$ and the $\max. \{|x|, |y|\} = 2$ is _____.

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RANK 44
V VERGIYA KAUSHAL
HT.No. 231004058



RANK 46
SUTHAR HARSHAL
HT.No. 231005060



RANK 58
ABHI JAIN
HT.No. 232091036



RANK 61
RIDAM JAIN
HT.No. 232068021



RANK 62
SUTHAR HARSH
HT.No. 232100028



RANK 63
SOJITRA DIPEN
HT.No. 231017001



RANK 65
D PANENNDRANADHA REDDY
HT.No. 236050288



RANK 67
GUNDA SUSHANTH
HT.No. 236129732



RANK 68
HARSH TAYA
HT.No. 237021178



RANK 69
KRISH GUPTA
HT.No. 232068189



RANK 73
KUNCHAKURTHY VIVEK
HT.No. 236134093



RANK 81
N DHARMA TEJA REDDY
HT.No. 236050138



RANK 84
S VENKAT KOUNDINYA
HT.No. 231030673



RANK 87
MRUNAL SHRICKANT
HT.No. 23104138



RANK 92
SAMOTA APURVA
HT.No. 232099210



RANK 97
N PRINCE BRANHAM REDDY
HT.No. 236064112

BELOW 20
All India Open Category Ranks

10
RANKS (50%)

BELOW 100
All India Open Category Ranks

32

BELOW 1000
All India Open Category Ranks

181

BELOW 100
All India Category Ranks Count

89

BELOW 1000
All India Category Ranks Count

699

NUMBER OF RANKS QUALIFIED
3,621

ADMISSIONS OPEN » JEE ADVANCED LONG-TERM 2024

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